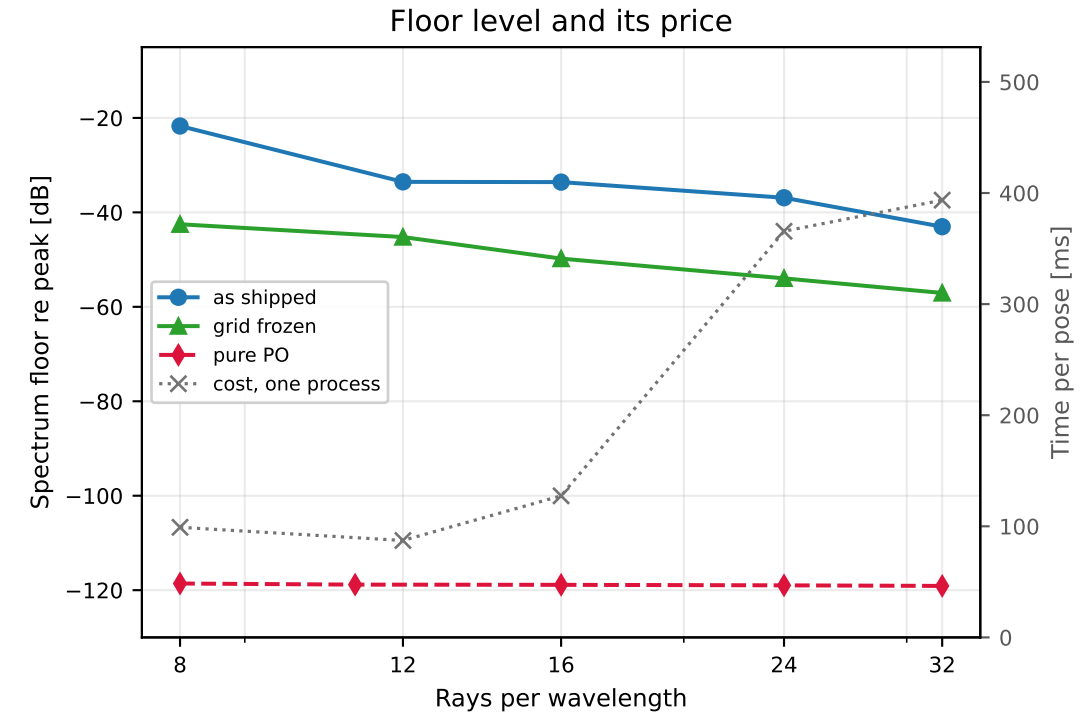
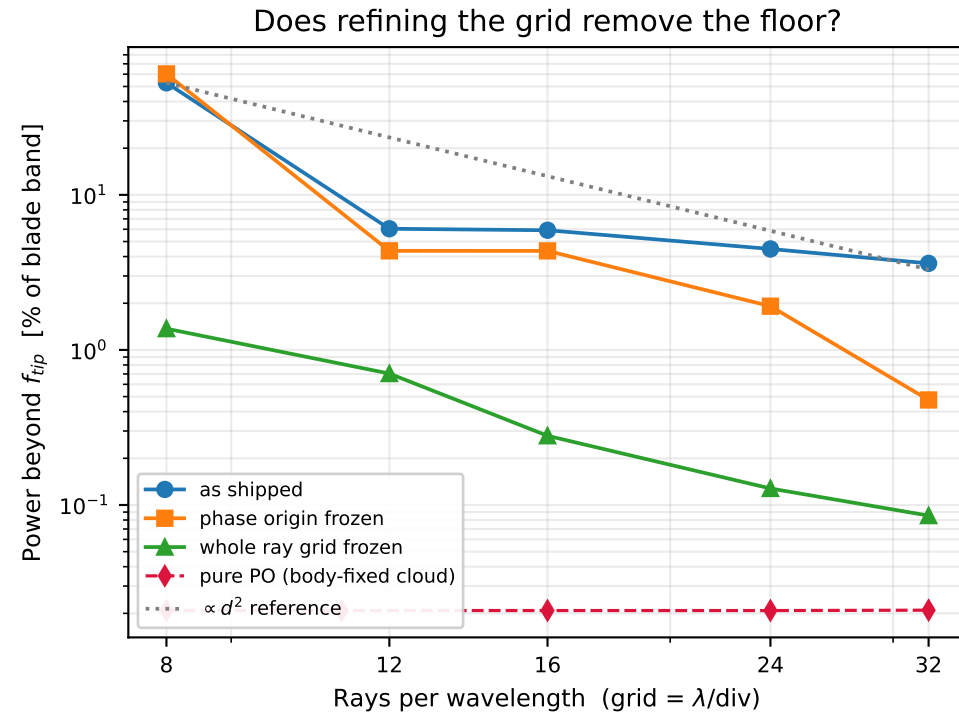
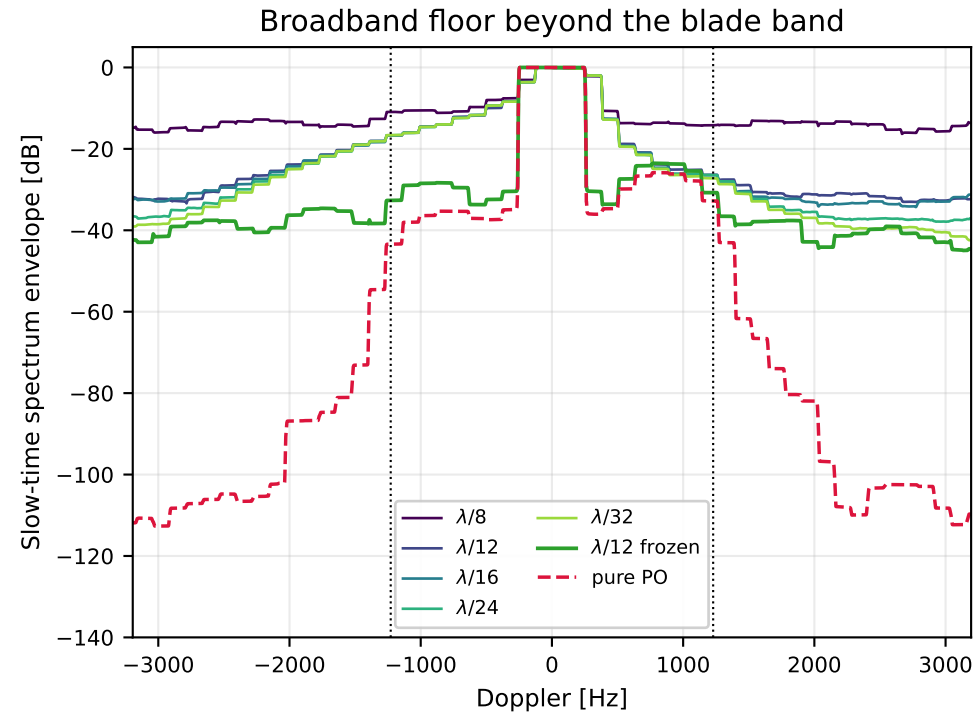
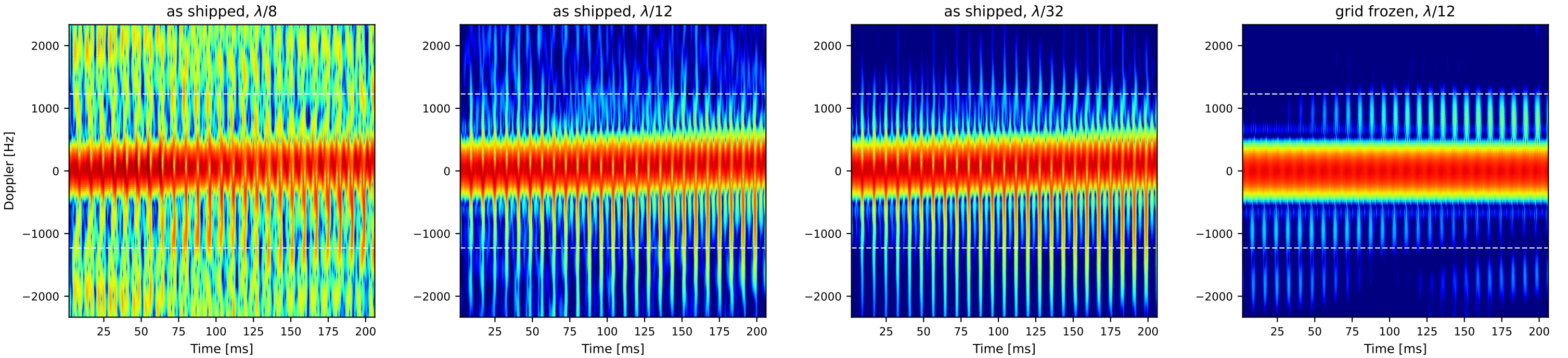


Is the out-of-band floor a ray-grid artefact?



Ray-grid convergence of the SBR micro-Doppler kernel. One axis only: the same DJI Matrice 4E, pose, rotor speeds, carrier and slow-time grid (4096 samples at 19700 Hz); only the ray spacing changes. The metric is the report-7b one, the fraction of blade-band power landing beyond the nominal tip Doppler 1229 Hz. As shipped, the grid is re-derived from each pose's bounding box, which breathes by 96 mm as the rotors turn, so every pose samples the target at an effectively random sub-cell offset; refining from $\lambda/12$ to $\lambda/32$ then buys only 2.2 dB (slope -0.55, where pure discretisation noise would give -2). Freezing the grid in the lab frame costs no extra rays, gains 9.3 dB at the production spacing, raises the in-band spectral agreement with the independent pure-PO engine from 0.44 to 0.95, and only then does the residual converge as d squared (slope -2.09, R^2 0.987). The pure PO control samples a point cloud attached to the body and is flat across the same ladder. Maps share one amplitude reference.